PERFECT COMPLEXES ON A SCHEME WITH THE RESOLUTION PROPERTY ARE THE LOCALIZATION OF THE HOMOTOPY CATEGORY OF FINITE LOCALLY FREE COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The perfect complexes form a saturated triangulated subcategory of the derived category, closed under two-out-of-three and direct summands. invoked at proof lines 10023-10027. Supplies the target side of the localization statement.
- For an exact functor out of a triangulated category, the quasi-isomorphisms in the source form a saturated multiplicative system compatible with the triangulated structure, the localization exists, and the functor factors through it universally. invoked at proof lines 10030-10042. Builds the localization S^{-1}K^b(A) and the exact factorization.
- The resolution property gives: every perfect complex is represented by a bounded complex of finite locally free modules, every map between such complexes in the derived category lifts to an actual map after resolution, and equal maps become homotopic after further resolution. invoked at proof lines 10043-10052. Shows the factorized functor is an equivalence.

Source reference. Original section 05RN: sources/stacks-chow/tags.txt, source lines 1–1. Referenced at statement lines 9994-9997 for the definition of K^b(A); the triangulated structure checks are inside the counted span.

Source reference. Original definition 05QM: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10018-10021 among the definitions used; the application to D_perf is inside the counted span.

Source reference. Original definition 05RB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10018-10021 among the definitions used; the verification that D_perf is saturated is inside the counted span.

Source reference. Original definition 05R2: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10018-10022 among the definitions used; the verification for S is inside the counted span.

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Source reference. Original definition 05Q8: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10018-10023 among the definitions used; the verification for S is inside the counted span.

Source reference. Original lemma 08CR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10024-10026 for the triangulated subcategory property; the localization assembly is inside the counted span.

Source reference. Original lemma 08CS: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10024-10027 for the saturated property; the localization assembly is inside the counted span.

Source reference. Original lemma 05RQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10027-10029 for K^b(A) being triangulated; the exactness of the functor is inside the counted span.

Source reference. Original lemma 05R4: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10031-10034 for S being a compatible saturated multiplicative system; the localization construction is inside the counted span.

Source reference. Original proposition 05R6: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10035-10038 for the existence of S^{-1}K^b(A); the factorization is inside the counted span.

Source reference. Original lemma 05R7: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10039-10042 for the exact factorization; the equivalence proof is inside the counted span.

Source reference. Original lemma 0F8F: sources/stacks-perfect/pinned-original-source.tex, source lines 9890–9890. Cited at proof lines 10043-10045 as the first equivalence input; the assembly of the three inputs is inside the counted span.

Source reference. Original lemma 0F8G: sources/stacks-perfect/pinned-original-source.tex, source lines 9912–9912. Cited at proof lines 10043-10046 as the second equivalence input; the assembly is inside the counted span.

Source reference. Original lemma 0F8H: sources/stacks-perfect/pinned-original-source.tex, source lines 9952–9952. Cited at proof lines 10043-10047 as the third equivalence input; the conclusion that the factorized functor is an equivalence is inside the counted span.

Source reference. Original lemma 05SQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10053-10057 to conclude the equivalence is one of triangulated categories; the final assembly is inside the counted span.

Proposition 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Denote*

1. *$\mathcal{A}$ the additive category of finite locally free $\mathcal{O}_X$-modules,*
2. *$K^b(\mathcal{A})$ the homotopy category of bounded complexes in $\mathcal{A}$, see Derived Categories, Section 05RN, and*
3. *$D_{perf}(\mathcal{O}_X)$ the strictly full, saturated, triangulated subcategory of $D(\mathcal{O}_X)$ consisting of perfect objects.*

With this notation the obvious functor

$$K^b(\mathcal{A}) \longrightarrow D_{perf}(\mathcal{O}_X)$$

is an exact functor of trianglated categories which factors through an equivalence $S^{-1}K^b(\mathcal{A}) \to D_{perf}(\mathcal{O}_X)$ *of triangulated categories where* S *is the saturated multiplicative system of quasi-isomorphisms in* $K^b(\mathcal{A})$*.*

Proof. If you can parse the statement of the proposition, then please skip this first paragraph. For some of the definitions used, please see Derived Categories, Definition 05QM (triangulated subcategory), Derived Categories, Definition 05RB (saturated triangulated subcategory), Derived Categories, Definition 05R2 (multiplicative system compatible with the triangulated structure), and Categories, Definition 05Q8 (saturated multiplicative system). Observe that $D_{perf}(\mathcal{O}_X)$ is a saturated triangulated subcategory of $D(\mathcal{O}_X)$ by Cohomology, Lemmas 08CR and 08CS. Also, note that $K^b(\mathcal{A})$ is a triangulated category, see Derived Categories, Lemma 05RQ.

It is clear that the functor sends distinguished triangles to distinguished triangles, i.e., is exact. Then S is a saturated multiplicative system compatible with the triangulated structure on $K^b(\mathcal{A})$ by Derived Categories, Lemma 05R4. Hence the localization $S^{-1}K^b(\mathcal{A})$ exists and is a triangulated category by Derived Categories, Proposition 05R6. We get an exact factorization $S^{-1}K^b(\mathcal{A}) \to D_{perf}(\mathcal{O}_X)$ by Derived Categories, Lemma 05R7. By Lemmas 0F8F, 0F8G, and 0F8H this functor is an equivalence. Then finally the functor $S^{-1}K^b(\mathcal{A}) \to D_{perf}(\mathcal{O}_X)$ is an equivalence of triangulated categories (in the sense that distinguished triangles correspond) by Derived Categories, Lemma 05SQ. □